J. Phys. Soc. Jpn. 2002.71:2832-2835.
Downloaded from journals.jps.jp by University of Edinburgh on 12/14/18. For personal use only.

Journal of the Physical Society of Japan
Vol. 71, No. 12, December, 2002, pp. 2832–2835
©2002 The Physical Society of Japan

Superconducting Condensation Energy and Pseudogap Formation in $La_{2-x}Sr_xCuO_4$: New Energy scale for Superconductivity

Naoki MOMONO, Toshiaki MATSUZAKI, Migaku ODA and Masayuki IDO

Department of Physics, Hokkaido University, Sapporo 060-0810, Japan

(Received June 24, 2002)

The electronic specific heat C_{el} was measured for $La_{2-x}Sr_xCuO_4$ to examine the superconducting condensation energy at $T = 0$, U_0. We reconfirmed that U_0 is markedly reduced even in slightly underdoped samples whose T_c's are still more than 30 K. The reduction of U_0 results from the suppression of the effective superconducting gap, in addition to that of the density of states at the Fermi level E_F, caused by the pseudogap formation in the normal state. The doping-level (p) dependence of U_0 as well as T_c can be explained over a wide p range by introducing an effective energy gap $\Delta_{eff} = \beta p \Delta_0$ ($\beta = 4.5$) instead of Δ_0, where Δ_0 is the maximum gap at $T \ll T_c$.

KEYWORDS: condensation energy, pseudogap, specific heat, $La_{2-x}Sr_xCuO_4$
DOI: 10.1143/JPSJ.71.2832

One of the puzzling features of high-T_c cuprates is that the superconducting (SC) condensation energy at $T = 0$, U_0, is exceedingly reduced even in slightly underdoped samples with sufficiently high T_c values. This issue has been considered a clue to understanding the mechanism of high-T_c superconductivity, and discussed in terms of a normal state gap, the evolution of an interplane phase coherence, the appearance of magnetic resonance in neutron scattering below T_c and the change of angle-resolved photoemission spectroscopy (ARPES) spectra at $T \leq T_c$.[1–5] Another puzzling feature is the eccentric relationship between T_c and the maximum gap Δ_0 in the underdoped region; Δ_0 continues to increase with the decrease of the doping level (p), whereas T_c decreases in such a region.[6–8] Recently, it has been pointed out for $Bi_2Sr_2CaCu_2O_{8+\delta}$ (Bi2212) and $La_{2-x}Sr_xCuO_4$ (La214) that T_c roughly scales with $\kappa p \Delta_0$ ($\kappa = 1.5$–1.7) instead of Δ_0, as was first predicted by Lee and Wen[9] theoretically, over a wide p-range including a slightly overdoped region.[10,11] Such an eccentric relation, $T_c \sim \kappa p \Delta_0$, which is in sharp contrast to the BCS result, has been discussed on the basis of the shrinkage of the Fermi surface caused by the formation of a pseudogap or spinon pairing in the normal state.[10,11]

The pseudogap structure with an energy scale of order $2\Delta_0$ has been confirmed to appear around E_F in the electron excitation spectrum by ARPES and tunneling spectroscopy studies on Bi2212.[12–16] The pseudogap with the energy scale of order $2\Delta_0$, referred to as a small pseudogap (SPG), has qualitatively the same symmetry as the energy gap in the SC state.[12–14] Recent tunneling spectroscopy measurements on Bi2212 have demonstrated that the SPG tends to open around the mean-field temperature T_{co} [$= 2\Delta_0/(4$–$5)k_B$] for d-wave superconductors.[17–21] The SPG was first reported as the spin gap in NMR relaxation time T_1 measurements on $YB_2C_3O_{6+\delta}$,[22] and has been found in NMR measurements of many high-T_c cuprates, except one typical member, La214.[23,24]

Photoemission spectroscopy studies on La214 and Bi2212 have demonstrated that in addition to the SPG, another type of pseudogap develops except in highly doped samples. Low-temperature ARPES measurements on La214 revealed that the flat bands near $(\pi, 0)$ and $(0, \pi)$, one of the striking features of the electronic structure in high-T_c cuprates, shift downward from the Fermi level E_F with the decrease of x at $x \lesssim 0.2$.[25–27] Since the flat bands make a large contribution to the density of states (DOS), the downward shift of the flat band leads to a marked reduction of the DOS around E_F. The energy region with the reduced DOS around E_F is often referred to as a large pseudogap (LPG). The LPG has an energy scale on the order of the effective antiferromagnetic (AFM) exchange energy J_{eff} ($>2\Delta_0$), and starts to develop around the temperature T_{max} ($\sim J_{eff}/k_B$) at which the magnetic susceptibility χ is maximal.[25–28] Since χ decreases gradually at $T \leq T_{max}$ through the development of an AFM correlation, the LPG is expected to originate from the formation of the AFM gap.

Very recently it has been reported in ARPES measurements on La214 that the Fermi surface shrinks to the so-called "nodal Fermi arcs" centered at the nodal points near $(\pi/2, \pi/2)$ in the lightly doped region, as in Bi2212.[29] In the underdoped region of Bi2212, the Fermi surface is truncated near $(\pi, 0)$ and $(0, \pi)$ owing to the SPG formation, and shrinks to the nodal arcs.[14]

In the present work, the electronic specific heat C_{el} of La214 was measured over a wide p-range and the condensation energy U_0 was evaluated from the result of C_{el}. In highly doped samples ($x > 0.2$), U_0 agreed with the BCS value, but it was exceedingly reduced in the underdoped region, as was reported by Loram *et al.*[1] The reduction of U_0 is too large to be explained by taking into account the reduction of $N(0)$ caused by the formation of pseudogaps only. We point out that a new energy scale $\Delta_{eff} = \beta p \Delta_0$ ($\beta = \text{const.}$) should be introduced instead of Δ_0 to explain the excessive reduction of U_0 in the pseudogap regime ($x < 0.2$).

Ceramic samples of La214 used for specific heat measurements were prepared by using a solid reaction in an oxygen atmosphere. The critical temperature T_c was determined from the diamagnetic transition curve measured by a SQUID magnetometer. Specific heat measurements were carried out using a conventional pulsed-heat technique. The electronic specific heat C_{el} was obtained by subtracting the phonon term C_{ph}, which was obtained from Ni-substituted non-superconducting samples, from the observed

J. Phys. Soc. Jpn. 2002.71:2832-2835.
Downloaded from journals.jps.jp by University of Edinburgh on 12/14/18. For personal use only.

specific heat of superconducting samples. The partial Ni-substitution for Cu makes it possible to determine the phonon term C_{ph} precisely. This is because the partial substitution of Ni has little effect on the phonon term C_{ph}, in contrast with the partial substitution of Zn, which modifies the phonon term C_{ph} significantly. Details of the measurements of C_{ph} have been published elsewhere.[30)]

Figure 1 shows typical results of C_{el} for $La_{2-x}Sr_xCuO_4$ with C_{el}/T vs T plots. The electronic specific heat C_{el} was obtained by using the phonon term C_{ph}^{Ni} of Ni-doped non-superconducting samples, as mentioned above. The C_{el}/T ($\equiv\gamma$) for $x \leq 0.2$ shows a small, broad peak at temperature $T' \sim T_{co}$ [$\equiv 2\Delta_0/(4\text{–}5)k_B$] and gradually decreases at $T < T'$, implying that the DOS at E_F, $N(0)$, will be suppressed due to the formation of an SPG at $T \lesssim T_{co}$.[31)] It should be noted that such SPG behavior appears in almost the same x-range ($x \lesssim 0.2$) as the LPG.[25,26)]

In the highly doped $x = 0.22$ sample exhibiting no pseudogap behavior, the anomaly of C_{el} below T_c is in good agreement, in both shape and size, with the BCS result for d-wave superconductors (the inset in Fig. 1). However, in the pseudogap regime ($x < 0.2$), the anomaly of C_{el} is different from the BCS result. In particular, the anomaly of C_{el} for $x = 0.1$ is significantly enhanced over a wide temperature range above T_c and steeply reduced below T_c, leaving a sharp peak at T_c. The sharp peak of C_{el} at T_c means that the enhancement of C_{el} at $T > T_c$ does not result from distribution in T_c due to any inhomogeneity effect, but from a strong fluctuation effect. As seen in Figs. 1 and 2, the fluctuation effect above T_c is more evident in samples with a low doping level. Hereafter, the T-range between T_c and the temperature T_{sf}, where the fluctuation effect becomes evident and the γ-value begins to increase, is referred to as the fluctuation region (Fig. 2). On the other hand, the steep reduction of C_{el} at $T < T_c$ indicates that the energy gap grows rapidly below T_c and quasiparticle excitations are largely suppressed. These features suggest that the SC order parameter develops to a large extent at $T > T_c$ but its phase fluctuates largely in the pseudogap regime.

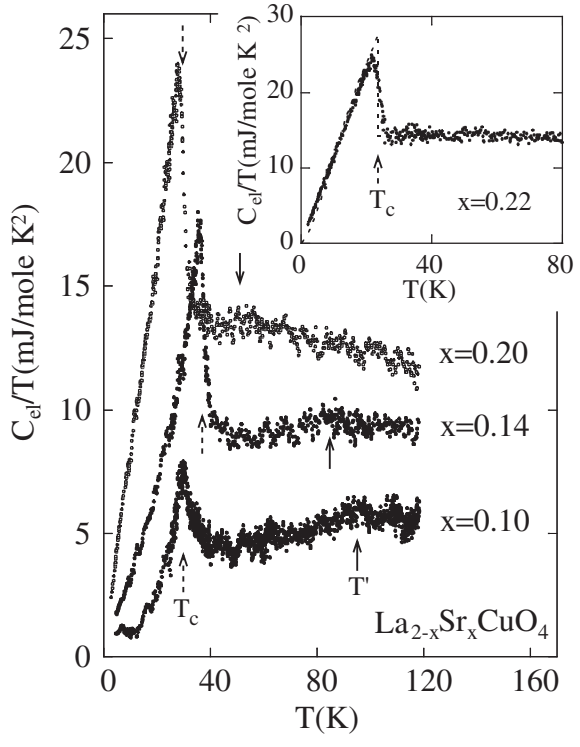


Fig. 1. Electronic specific heat C_{el} for La214 ($0.1 \leq x \leq 0.2$) plotted with C_{el}/T vs T. The inset shows the C_{el}/T vs T curve for $x = 0.22$. The dotted line in the inset represents the theoretical result for d-wave BCS superconductors.

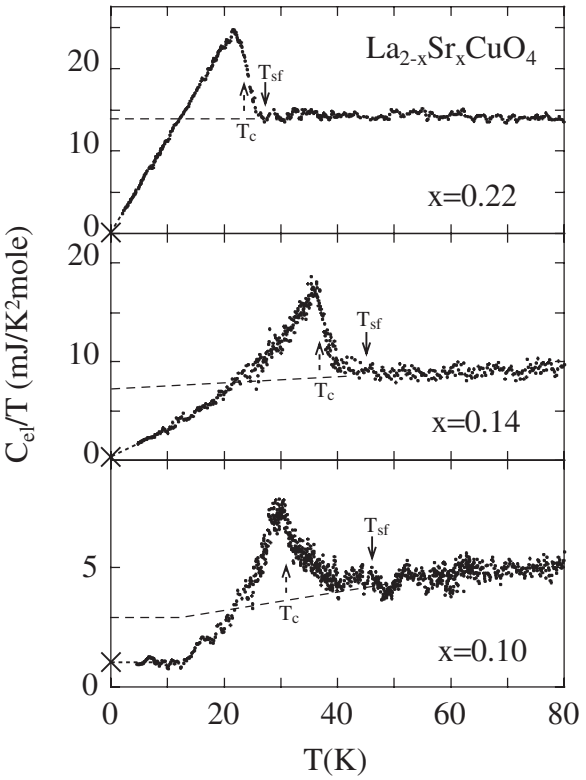


Fig. 2. T-dependence of C_{el}/T ($=\gamma$) in the superconducting state, $\gamma_s(T)$, and the normal state, $\gamma_n(T)$, for La214. The broken line represents $\gamma_n(T)$ at $T < T_{sf}$, where T_{sf} is the temperature around which the fluctuation effect becomes evident. The residual γ-value (γ_0) at $T = 0$ in the superconducting state, which was obtained in C_{el}/T vs T^2 plots, is also plotted on the vertical axis ($\times$).

We can evaluate the SC condensation energy U_0 by integrating the entropy difference $S_n - S_s$ between $T = 0$ and T_{sf},

$$U_0 = \int_0^{T_{sf}} (S_n - S_s)\mathrm{d}T, \tag{1}$$

where the subscripts "s" and "n" stand for the SC and the normal state, respectively. Given both γ_s and γ_n, we can obtain the entropy S_s and S_n by executing the integration

$$S_{s,n}(T) = \int_0^T \left(\frac{C_{el}}{T}\right)_{s,n} \mathrm{d}T = \int_0^T \gamma_{s,n}\mathrm{d}T,$$

and evaluate the condensation energy U_0. In the present study, to obtain γ_n for $T < T_{sf}$, γ_n ($T < T_{sf}$), we extrapolated the data of γ_n ($T > T_{sf}$) down to below T_{sf}, as shown for typical samples in Fig. 2. In highly doped $x = 0.22$ samples, since γ_n ($T > T_{sf}$) is constant over a wide T-range above T_c, we can safely extrapolate the data down below T_{sf} on the assumption that $\gamma_n = \text{const}$ at $T < T_{sf}$. The γ_n ($T < T_{sf}$) thus obtained satisfies the constraint on the second order phase transition $S_n(T_{sf}) = S_s(T_{sf})$. This constraint is often called

J. Phys. Soc. Jpn. 2002.71:2832-2835.
Downloaded from journals.jps.jp by University of Edinburgh on 12/14/18. For personal use only.

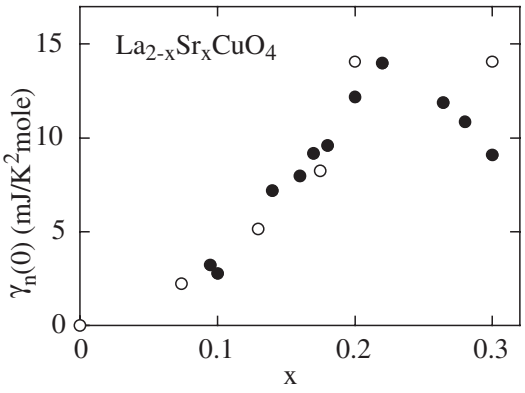


Fig. 3. Doping-level dependence of $\gamma_n(0)$ values for La214. The $\gamma_n(0)$ for non-superconducting samples with $x > 0.26$ were obtained from the conventional C/T vs T^2 plots. The open circles represent the density of states at E_F, $N(0)$, measured by angle-integrated photoemission spectroscopy on La214.[22] $N(0)$ is normalized with $\gamma_n(0)$ at $x = 0.22$.

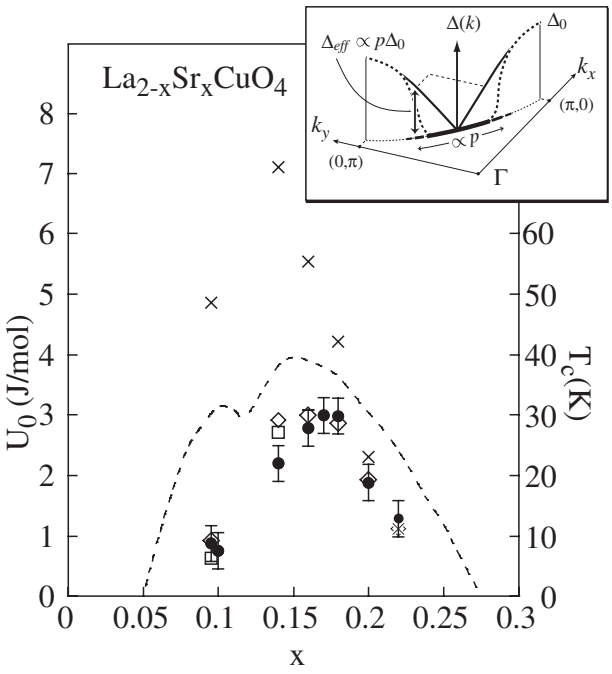


Fig. 4. Superconducting condensation energy U_0 for La214. Experimental data (•) obtained by executing the integral of eq. (1) in the text are shown, together with the calculated values (×) given by eq. (2) for $\gamma_n(0)$ and Δ_0. The calculated values (◇) obtained by substituting Δ_{eff} ($= \beta p \Delta_0$) for Δ_0 in eq. (2) are also shown, where the values for $x \leq 0.14$ were calculated with $\gamma_n(0) - \gamma_o$ (□) as well as $\gamma_n(0)$. The inset shows schematic illustration of the SPG at $\gtrsim T_c$ (dotted line) and the nodal Fermi arc (bold solid line). At $T < T_c$, the SC gap develops on the nodal Fermi arc (solid line).

"the entropy balance", because the excess and deficit parts of the γ_s vs T curve, compared with γ_n ($T \lesssim T_{sf}$), must be equal in area to satisfy the condition $S_n(T_{sf}) = S_s(T_{sf})$. On the other hand, γ_n ($T > T_{sf}$) for $x \leq 0.2$ becomes temperature dependent due to the formation of the SPG, so we extrapolated the data of γ_n ($T > T_{sf}$) down to below T_{sf} using a declining straight line to satisfy the entropy balance. Furthermore, in samples for $x \leq 0.1$, we assumed $\gamma_n = \text{const}$ for $T < 0.3T_c$ because the γ_s value tends to be saturated below $\sim 0.3T_c$ in these samples as shown in Fig. 2.

We calculated S_s and S_n using γ_s and γ_n ($T < T_{sf}$), and evaluated the SC condensation energy U_0 by executing the integral of eq. (1) over the SC temperature range, including the fluctuation region. The present condensation energy, plotted against x ($= p$) in Fig. 4., is consistent with the results reported by Loram *et al.* for $x \gtrsim 0.2$, but is larger by about 40% for $x < 0.2$.[1] The SC condensation energy U_0 (eV) for a flat DOS is

$$U_0 = \alpha N(0)\Delta_0^2/2 \approx 2.1 \times 10^{-5}\alpha\gamma_n(0)\Delta_0^2/2 \qquad (2)$$

for d-wave BCS superconductors ($\alpha \sim 0.4$), where $\gamma_n(0)$ represents the value of γ_n at $T = 0$.[32,33] In Fig. 4, U_0 is the condensation energy given by eq. (2) for $\gamma_n(0)$ (J/K^2 mole) and Δ_0 (eV) determined in tunneling experiments on La214 except the $x = 0.22$ sample.[8,34] For the $x = 0.22$ sample, whose superconducting anomaly of C_{el} is well described by the BCS theory as mentioned above, Δ_0 was estimated from T_c using the BCS relation $2\Delta_0 = 4.3k_BT_c$ for d-wave superconductors. In samples for $x \lesssim 0.14$, since γ_s ($T = 0$) determined in C_s/T vs T^2 plots has a small residual value (γ_o) in the SC state, we used $\gamma_n'(0) \equiv \gamma_n(0) - \gamma_o$ as well as $\gamma_n(0)$ in the evaluation of U_0. The condensation energy U_0 given by eq. (2) is in good agreement with the experimental value for the overdoped $x = 0.22$ sample whose SC properties are of the d-wave BCS type. However, the calculated values of U_0 for $x \lesssim 0.2$ are quite different from the experimental results, though the marked reduction of U_0 in the underdoped region is qualitatively reproduced in the calculation through the reduction of $\gamma_n(0)$, i.e., $N(0)$. The serious disagreement between experimental and calculated values of U_0 at $x \lesssim 0.2$ seems to originate from the modification of the effective SC energy gap caused by the SPG formation. In fact, it has been pointed out that T_c roughly scales with $p\Delta_0$ instead of Δ_0 in the pseudogap regimes of Bi2212 and La214, whereas T_c scales with Δ_0 in highly doped samples which exhibit no pseudogap.[10,11] This implies that the effective SC gap, Δ_{eff}, will not be Δ_0 but $\beta p\Delta_0$ ($\beta = \text{const.}$) in the pseudogap regime. Then we calculate U_0 again by substituting $\beta p\Delta_0$ for Δ_0 in eq. (2) and treating β as a fitting parameter. In Fig. 4, the calculated values of U_0 for $\beta = 4.5$ are plotted against x ($= p$). The new results reproduce the experimental values very well over the entire p-range examined, as seen in Fig. 4. The new calculated result can explain the experimental value even for $x = 0.22$, where Δ_0 should be used, instead of Δ_{eff}, in the calculation of U_0 because no pseudogap develops there. This paradox is, however, dissolved by noting the fact that factor β ($= 4.5$) in $\Delta_{eff} = \beta p\Delta_0$ gives $\beta p \sim 1$ for p ($= x$) $= 0.22$, namely, Δ_{eff} reduces to Δ_0 at $x = 0.22$.

As pointed out by many researchers, mobile carriers on the nodal parts of the Fermi surface near $(\pi/2, \pi/2)$ will have much higher mobility within the Cu–O plane than those on antinodal parts near $(\pi, 0)$ and $(0, \pi)$.[9,35–37] This is typically demonstrated by the following observation: the Fermi surface is truncated near antinodal points through the evolution of the SPG at $T > T_c$ and shrinks into the nodal arcs, but the in-plane resistivity tends to be slightly suppressed rather than enhanced.[38] Since mobile carriers with high mobility will play a crucial role in making the collective motion of the Cooper pairs coherent when they form Cooper pairs and participate in the collective motion of the pairs,[35,36] the nodal Fermi arcs are expected to drive the

J. Phys. Soc. Jpn. 2002.71:2832-2835.
Downloaded from journals.jps.jp by University of Edinburgh on 12/14/18. For personal use only.

superconductivity and dominate the SC properties such as U_0 and T_c. This enables us to speculate that in the pseudogap regime of high-T_c cuprates, the energy gap formed over the nodal Fermi arcs will play a role as an effective SC gap, though the energy gap with a *d*-wave symmetry is formed over the whole Fermi surface below T_c with the maximum gap Δ_0 at antinodal points. Such a speculation is consistent with experimental results of Raman scattering measurements showing that the B_{1g} Raman continua weighing out antinodal parts of the Fermi surface are indicative of a marked suppression of the coherent pairing gap in the underdoped region, whereas the B_{2g} continua weighing out nodal parts indicate the existence of the coherent energy gap that scales with T_c.[39,40]

Wen and Lee have proposed on the basis of the SU(2) slave-boson model that a hole pocket will appear near $(\pi/2, \pi/2)$ due to spinon pairing in the spin gap (pseudogap) regime.[41] The hole pocket will extend towards $(\pi, 0)$ and $(0, \pi)$ with a length of $\sim p$. However, such a hole pocket has not been confirmed experimentally; instead of the hole pocket, the Fermi surface changes to the nodal arcs in the pseudogap regime, as mentioned above. The length of the Fermi arc is difficult to estimate precisely because it smears at its edges,[29] but if its effective length is proportional to $\sim p$ $(= x)$ at $T \sim T_c$, the situation becomes analogous to that in the model of Lee and Wen. In this case, the gap size at the edges of the Fermi arcs, corresponding to the hole pocket in the Lee and Wen model, is roughly proportional to $p\Delta_0$ at $T \ll T_c$ as long as the linear dimension of the Fermi arc is not so large (the inset in the Fig. 4), as pointed out by Lee and Wen. This gap size is consistent with that of Δ_{eff} introduced in the analysis of U_0.

We measured the electronic specific heat C_{el} of La214 over wide *p*- and *T*-ranges, and evaluated the SC condensation energy U_0 from the result of C_{el}. We reconfirmed that U_0 and $\gamma_n(0)$, i.e., $N(0)$, are markedly suppressed in the pseudogap regime ($x \lesssim 0.2$). The reduction of U_0 is too large to be explained by taking into account the reduction of $\gamma_n(0)$ only. To explain the experimental values of U_0 in the pseudogap regime, we have to introduce a new energy scale Δ_{eff} $(= \beta p \Delta_0)$ instead of Δ_0. Presumably, the energy gap formed over the nodal Fermi arcs near $(\pi/2, \pi/2)$ will play a role as the effective SC gap.

This work was supported in part by Grants-in-Aid for Scientific Research on Priority Area (Novel Quantum Phenomena in Transition Metal Oxides) and on Projects (Nos. 12740191 and 13440104) from the Ministry of Education, Culture, Sports, Science and Technology of Japan.

1) J. W. Loram *et al.*: J. Supercond. **7** (1994) 243.
2) P. W. Anderson: Science **279** (1998) 1196.
3) D. J. Scalapino and S. White: Phys. Rev. B **58** (1998) 8222.
4) E. Demler and S.-C. Zhang: Nature **396** (1998) 733.
5) M. R. Norman *et al.*: Phys. Rev. B **61** (2000) 14742.
6) Ch. Renner *et al.*: J. Low Temp. Phys. **105** (1996) 1073.
7) J. M. Harris: Phys. Rev. B **54** (1996) 15665.
8) T. Nakano *et al.*: J. Phys. Soc. Jpn **67** (1998) 2622.
9) P. A. Lee and X. G. Wen: Phys. Rev. Lett. **78** (1997) 4111.
10) M. Ido, N. Momono and M. Oda: J. Low Temp. Phys. **117** (1999) 329.
11) M. Oda *et al.*: J. Phys. Soc. Jpn. **69** (2000) 983.
12) A. G. Loeser *et al.*: Science **273** (1996) 325.
13) H. Ding *et al.*: Nature **382** (1996) 51.
14) M. R. Norman *et al.*: Nature **392** (1998) 157.
15) Ch. Renner *et al.*: Phys. Rev. Lett. **80** (1998) 149.
16) M. Suzuki, S. Karimoto and K. Namekawa: J. Phys. Soc. Jpn. **67** (1998) 732.
17) M. Oda *et al.*: J. Phys. Soc. Jpn. **71** (2002) 1535.
18) T. Tanamoto *et al.*: J. Phys. Soc. Jpn. **61** (1992) 1886.
19) Y. Yanase and K. Yamada: J. Phys. Soc. Jpn. **68** (1999) 2999.
20) Q. J. Chen *et al.*: Phys. Rev. Lett. **81** (1998) 4708.
21) S. S. Lee and S.-H S Salk: Phys. Rev. B **64** (2001) 052501.
22) H. Yasuoka *et al.*: *Strong Correlation and Superconductivity* (Springer-Verlag, New York, 1989) Springer Series in Solid State Science Vol. 89, p. 254.
23) R. E. Walstedt, R. F. Bell and D. B. Mitzi: Phys. Rev. B **44** (1991) 7760.
24) K. Ishida *et al.*: J. Phys. Soc. Jpn. **63** (1994) 1104.
25) A. Ino *et al.*: Phys. Rev. Lett **81** (1998) 2124.
26) A. Ino *et al.*: Phys. Rev. B **65** (2002) 94504.
27) T. Sato *et al.*: Physica C **341–348** (2000) 815.
28) T. Nakano *et al.*: Phys. Rev. B **49** (1994) 16000.
29) T. Yoshida *et al.*: cond-mat/0206496.
30) T. Nagata *et al.*: Physica C **264–365** (2001) 430.
31) N. Momono *et al.*: J. Low Temp. Phys. **117** (1999) 353.
32) F. J. Ohkawa: J. Phys. Soc. Jpn. **56** (1987) 2267.
33) Y. Sun and K. Maki: Phys. Rev. B **51** (1995) 6059.
34) In the present study, the maximum gap Δ_0 for $x = 0.095$, which was determined from the tunneling spectra measured by using the point contact junction technique by M. Oda, L. Ozyuzer, J. F. Zasadzinski and K. E. Gray, was used, although the tunneling data have not been published.
35) D. Pines: Physica C **282–287** (1997) 273.
36) V. B. Geshkenbein, L. B. Ioffe and A. I. Larkin: Phys. Rev. B **55** (1997) 3173.
37) N. Furukawa, T. M. Rice and M. Salmhofer: Phys. Rev. Lett. **81** (1998) 3195.
38) Y. Yanase and K. Yamada: J. Phys. Soc. Jpn. **68** (1999) 548.
39) J. G. Naeini *et al.*: Phys. Rev. B **59** (1999) 9642.
40) M. Opel *et al.*: Phys. Rev. B **61** (2000) 9752.
41) X. G. Wen and P. A. Lee: Phys. Rev. Lett. **80** (1998) 2193.